

HASSAN SAEED HASSAN ALBATTRA

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SUMMARY

DevOps Engineer with MSc-level training in cloud computing and AI-native software engineering, focused on cloud infrastructure, Continuous Integration/Continuous Deployment (CI/CD) automation, containerized deployments, and security-first pipeline design. Hands-on experience provisioning Amazon Web Services (AWS) infrastructure with Terraform, deploying containerized Machine Learning (ML) models on EC2, and building end-to-end cloud pipelines using S3, EMR, VPC, and IAM. Completing AWS DevOps professional training toward AWS Developer and DevOps Engineer Associate certifications.

EDUCATION

Queen's University, Kingston, ON, Canada Jan. 2026 – Present
MSc – School of computing(SD-AI ERA).

Focus Areas: DevOps, MLOps, CI/CD, GitOps, Kubernetes, Infrastructure as Code, Cloud Computing, Generative AI, LLM applications.

Fayoum University, Fayoum, Egypt Sep. 2023 – Sep. 2026
MSc – Electrical Engineering (Computer & Systems Engineering)

Thesis: Phishing URL detection using ML/DL — built a Dockerized, ONNX-optimized inference pipeline meeting real-time SLAs. Published open-source code (GitHub) and dataset (Zenodo, 339K URLs).

Misr University for Science and Technology (MUST), Giza, Egypt Sep. 2019 – Feb. 2023
BEng – Electrical Engineering (Computer & Software Engineering) · Ranked 1st in Class

EXPERIENCE

Misr University for Science and Technology (MUST) Feb. 2023 – Present
Teaching Assistant – Computer & Software Engineering Dept. Giza, Egypt

- Delivered lectures and hands-on labs for 300+ students/semester.
- Designed 15+ hands-on labs on encryption algorithms, secure coding, ethical hacking, and threat modeling; supervised 10+ capstone projects on secure system deployments.
- Mentored students on Git workflows, automated testing, and secure SDLC practices; 3 students presented security research at departmental symposia.

PROJECTS

MLOps Qwen2.5 – A/B Testing & Drift Monitoring 2026

- Built an end-to-end MLOps pipeline for **Qwen2.5-1.5B**: tracked few-shot and **GPTQ 4-bit** quantization experiments with **MLflow**, achieving a 3.6× latency speedup and 65% size reduction.
- Served models via **vLLM** with an **80/20 A/B testing FastAPI router**, implemented drift monitoring via **Evidently**, and orchestrated automated canary deployments in **k3d** using **Argo Rollouts** gated by **Prometheus** metrics. [\[GitHub\]](#)

TestSeed – AI-Powered MongoDB Seed Data Generator 2026

- Built a full-stack TypeScript web application that generates realistic, schema-aware MongoDB seed data from Mongoose schemas or live MongoDB schema discovery.
- Implemented feedback-based regeneration, JSON/JavaScript seed export, direct MongoDB seeding, **seedBatchId** tracking, insertion reports, and rollback support.
- Used Next.js, React, Tailwind CSS, Express API, MongoDB/Mongoose, OpenAI API, and validation logic to preserve required fields, enums, unique constraints, and ObjectId references.
- Collaborated in a 4-member team using AI-assisted SDLC tools including Spec Kit, Cursor, Claude, GitHub Copilot, and GitHub Issues. [\[GitHub\]](#)

DevOps Sentiment K8s GitOps – CI/CD & GitOps Pipeline 2026

- Built a complete **Gitea Actions CI pipeline** for a FastAPI/TextBlob ML microservice: automated linting (**ruff**), static type checking (**mypy**), pytest with coverage, **multi-stage Docker build** (non-root, python:3.11-slim), image push to private registry, and **SBOM generation** (CycloneDX).
- Deployed to **k3d Kubernetes** via **ArgoCD GitOps** with canary release strategy; configured **Prometheus + Grafana** observability dashboard and OpenTelemetry instrumentation. [\[GitHub\]](#)

Cyber-SOC-Chatbot – End-to-End Cloud LLM on AWS 2026

- Provisioned full AWS infrastructure with **Terraform** (VPC, public/private subnets, IGW, security groups, IAM roles, S3, EMR) following the Well-Architected Framework.
- Preprocessed 20M+ cybersecurity events using **PySpark on AWS EMR**; fine-tuned TinyLlama-1.1B with **QLoRA/PEFT**, exported to GGUF, deployed via **Ollama on EC2 g4dn.xlarge** with systemd.
- Served model through **OpenWebUI (Docker)** with browser chat; configured security groups for SSH, API, and UI. [\[GitHub\]](#)

EdgePhish-5G – Dockerized ML Security Pipeline

2024 – 2025

- Containerized (Docker) a hybrid BERT–TF-IDF phishing detection pipeline; ONNX Runtime optimization reduced P99 latency from 17.2 ms to 7.3 ms (2.36×), meeting ≤ 10 ms SLA.
- Implemented reproducible, single-command deployment with environment-variable config; published on [\[GitHub\]](#) and [\[Zenodo\]](#) (339K URLs).

GuardedRAG – Safe Retrieval-Augmented Generation System

2026

- Built a secure RAG pipeline: PDF ingestion, LangChain chunking, Sentence Transformer embeddings, and semantic retrieval from **ChromaDB**.
- Orchestrated **Gemini 2.5 Flash** + **OpenAI Moderation API** in a 5-stage safety pipeline (scope check → moderation → retrieval → generation → output check); deployed via Gradio. [\[GitHub\]](#) · [\[Live Demo\]](#)

Crime Type Classification – ML for Police Resource Allocation

2026

- Built a leakage-aware, chronological ML pipeline (241K Toronto crime records, 393 features) comparing 8 model configurations; XGBoost with inverse-frequency class weighting achieved best macro F1 (0.426) across 5 crime types under severe class imbalance.
- Engineered cyclical temporal features, one-hot spatial encoding, and evaluated SMOTE resampling and MLP baselines with GridSearchCV + TimeSeriesSplit. [\[GitHub\]](#)

TECHNICAL SKILLS AND TRAINING

Languages: Python, Bash/Shell, Java, C/C++, JavaScript, SQL, YAML/JSON

Cloud / DevOps: Amazon Web Services (AWS), Docker, Kubernetes, Terraform, CloudFormation, Ansible, Continuous Integration/Continuous Deployment (CI/CD), GitHub Actions, Jenkins, ArgoCD, GitOps, Linux, Infrastructure as Code (IaC)

AWS Services: EC2, S3, EMR, Lambda, API Gateway, DynamoDB, RDS, VPC, IAM, ECS, EKS, SNS, SQS, CloudWatch, CloudTrail, Cognito, CodePipeline, CodeBuild, Systems Manager, Elastic Beanstalk, X-Ray

DevSecOps: Secure pipeline design, secrets management, IAM least privilege, encryption, Multi-Factor Authentication (MFA), threat modeling

AI / ML Deployment: PyTorch, ONNX Runtime, scikit-learn, XGBoost, Hugging Face, QLoRA/PEFT, Ollama, GGUF, Retrieval-Augmented Generation (RAG), LangChain, Gradio

Monitoring / MLOps: Prometheus, Grafana, MLflow, OpenTelemetry, vLLM, DVC, model versioning

Databases / Tools: PostgreSQL, MongoDB, ChromaDB, Git, GitHub, Jupyter, VS Code

Certifications (in progress): AWS Developer Associate · AWS DevOps Engineer Associate

Training: AWS DevOps Track (180 hrs, NTI) — Cloud Practitioner · Cloud Developer · Cloud DevOps | Offensive Cybersecurity (B.S.A) · Cybersecurity II (Security Meter) · Embedded Systems 80 hrs (AMIT) · Web Development 60 hrs (New Horizons)

PUBLICATION

H. AlBattar, R. Abul Seoud, D. A. Salem, “EdgePhish-5G: Hybrid BERT–TF-IDF Phishing URL Detection for Real-Time 5G Edge Deployment,” *Accepted to ITC*, 2026. [\[Dataset DOI\]](#), [\[Dataset DOI\]](#)